
UnLense: Unsupervised Super-Resolution of Gravitational Lensing Images via a Differentiable Lensing Forward Model

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Abstract

Strong gravitational lensing probes dark matter substructure directly, but upcoming surveys will find of order 10^5 galaxy-galaxy lenses and follow up only a small fraction, so the image pairs supervised super-resolution requires will not exist for most of that population. We present UnLense, which recovers a lens and a source from a single observation using the forward model as its only supervision: a differentiable elliptical power-law deflection field, a measured instrument response and detector binning render any proposed source into a predicted image, and a χ^2 against the observed pixels is the only training signal. Physics fixes the forward model but not the source representation, so we measure it on 2,000 held-out Roman-like simulations across four representations that share one lens, one optimizer and one scorer. Source error falls monotonically as the number of free source values falls, from nmse 0.056 at 3,844 free pixels to 0.017 at seven analytic parameters, a ceiling that exists only because these sources are analytic by construction. UnLense recovers seven physical parameters from the pixels alone at Spearman $+0.59$ to $+0.97$ and reproduces magnification quantities that never entered the fit. Of structure injected below the detector pixel scale a median 18 percent returns, and gating the source prior on local magnification, so that sub-pixel structure is permitted only where the lens delivers the resolution to constrain it, improves every source-plane metric on both seeds while the median χ^2 shifts by 0.15 percent.

1 Introduction

Gravitational lensing bends light from a background galaxy into arcs, rings and multiple images [1]. The deflection depends on the total projected mass and not on the light, so a lensed image carries information about the small-scale dark matter substructure that separates competing dark matter models [2, 3]. Lensing also magnifies, spreading a small patch of the source across many detector pixels, so super-resolution here is something the geometry already permits.

Recovering the source means inverting a map that is many-to-one by construction. The usual machine learning answer is supervised super-resolution on paired low- and high-resolution images, which has been done for lensing with a conditional diffusion model trained on matched ground- and space-based cutouts [19]. Those pairs do not exist at the scale required: that dataset ran to 173 real objects, while Euclid, LSST and Roman are expected to find of order 10^5 galaxy-galaxy lenses [4] and follow-up will reach a small fraction of them.

An alternative is to use the forward model as the supervision: a candidate lens and source are ray-traced, convolved with the instrument response, binned onto detector pixels and compared with the

observation, so no high-resolution target is needed [5]. This does not settle what the source should be. The deflection field is a small parametric family the arcs identify well, while published choices for the source run from a handful of analytic parameters [6, 7] through regularized pixel grids [14, 15] to tens of thousands of free pixels [8, 9], usually by convention rather than by measurement. Joint lens-and-source inference under a learned score-based prior has been demonstrated [16, 17]; we propose no new prior but measure what the choice is worth. We present `UnLense`, a label-free system that fits a lens and a source jointly to each observation, and use it to turn that convention into a measurement: data, deflection model, instrument model and scoring function are held fixed while only the source representation varies. We then test whether the recovered sub-pixel detail is real by injecting structure smaller than a detector pixel.

2 Data and methods

Dataset. Simulated galaxy-galaxy strong lensing images produced with `lenstronomy` [10] to resemble Roman Space Telescope observations, 127×127 detector pixels at 0.10593 arcsec per pixel, in three dark matter classes (axion, cold and warm) contributing equally, 10,000 for training and 2,000 held out, from disjoint splits. Each system pairs an elliptical power-law deflector with external shear, contributing no light of its own, with an elliptical Sérsic source [6] and a population of dark matter subhalos, and stores the noiseless unlensed source and the simulation parameters, so both can be scored against truth. That stored source is at detector sampling and already convolved, so no high-resolution target exists in the dataset.

Forward model. The lens equation is $\beta = \theta - \alpha(\theta)$, with α the deflection of an elliptical power-law convergence plus a constant external shear: Einstein radius θ_E , radial slope γ , two ellipticity and two shear components. This is the standard galaxy-scale mass model [11] and also the family the simulation used, so lens recovery here is a best case. The closed form of [11] calls `scipy.special.hyp2f1`, which has no PyTorch equivalent, so we evaluate the hypergeometric series it sums and reproduce `lenstronomy`'s deflection to machine precision. Rays are traced back from each image pixel to the source plane, the source is sampled there, the sky is convolved with the response and area-averaged onto detector pixels, in that order, because the optics blur before the detector integrates (Figure 1). The response is measured rather than assumed: the stored source is the analytic source already convolved with the true kernel, so the kernel is over-determined and a Wiener-regularized division in Fourier space stacked over 40 sources recovers it without assuming a shape (Appendix C).

Objective and initialization. The objective is $\chi^2 = \sum_i ((m_i - d_i)/\sigma)^2$ over a 45 pixel disk holding 6,361 pixels, with σ measured per image from a source-free annulus beyond radius 50. Runs that train a network replace σ^2 by $\sigma^2 + (0.02 m_i)^2$, a variance floor stating that the model is not trusted to better than two percent of its own value at any pixel; without it four images in every four hundred carry half of the batch gradient, because on the brightest arcs a one percent kernel-shape error is already a 50σ residual. Every χ^2 we report uses the plain σ , so the floor cannot flatter a result. No labels, no high-resolution target and no simulation parameters enter the objective, and starting values are measured from the image. Six lens parameters are used because, holding the true source fixed, arc-weighted correlation with the truth runs 0.799 for a circular isothermal sphere, 0.966 for an isothermal ellipsoid, 0.980 once the radial slope is freed and 0.996 for the ellipsoid with external shear.

Source representations. The lens model is settled by that measurement; the source is not, so we compare the five representations of Table 1 under one forward model, optimizer and scorer. Two are free pixel grids: one at twice detector sampling from a convolutional upsampling network [12, 13] acting on the observation back-projected into the source plane, one from the earlier grid pipeline under a total-variation prior. Two are hybrids in which a fitted Sérsic carries the wings and the network adds a bounded residual, decoded either convolutionally or from a pooled vector. The fifth is the Sérsic alone. Every network row uses a lens fitted per image and then frozen, so rows 2–5 share a common lens.

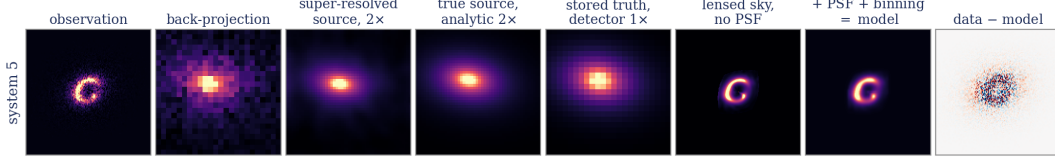


Figure 1: UnLense end to end. The observation is back-projected into the source plane, the network returns a source twice as finely sampled as the detector, and that source is re-lensed, convolved and binned to close the χ^2 against the same pixels. The only stored truth is at detector sampling and already convolved, so the analytic re-render is for the eye only.

Table 1: Five source representations scored on the same 2,000 validation images against the stored source, convolution matched. “Data per value” uses the 3,878 pixels above 3σ in the fit disk. Rows 2–5 are the controlled comparison, identical in lens, data, optimizer and scorer; row 1 is the earlier pipeline, which also used a fixed circular lens and a total-variation prior, so it bounds rather than isolates the cost of a free field.

source representation	free values	data/value	corr	size ratio	nmse
free field, 254^2 , circular lens, TV prior	64,516	0.06	–	10.6–12.8	–
free field, 62^2 , convolutional network	3,844	1.01	0.971	1.141	0.0560
Sérsic + 48^2 residual, same network	7 + 2,304	1.68	0.983	1.048	0.0336
Sérsic + 32^2 correction, pooled vector	7 + 1,024	3.76	0.985	1.026	0.0297
Sérsic only, 7 values	7	554	0.992	1.061	0.0166

3 Results

Parameter and source recovery. On 2,000 held-out images the seven simulation parameters are recovered at Spearman rank correlation $+0.973$ (θ_E), $+0.956$ (Sérsic index), $+0.947$ (half-light radius), $+0.925$ (source offset), $+0.813$ (lens ellipticity), $+0.645$ (external shear) and $+0.590$ (radial slope); none was available to the fit. Rank correlation is insensitive to a monotone bias, so median absolute errors accompany it in Appendix B. The recovered source matches the stored truth at correlation 0.992, size ratio 1.061 and nmse 0.017, against a size ratio of 10.6 to 12.8 for the earlier grid pipeline on these data.

Validation on quantities never fit. Magnification is derived from the recovered lens and enters no objective. Total magnification is 5.83 against a true 5.89, the tangential critical curve is recovered to a radial root mean square error of 0.56 detector pixels, and its azimuthal swing is 0.512 against 0.554 arcsec (Figure 2). A circular lens gives a swing of exactly zero, so recovering it shows the four shape parameters were recovered jointly, not in magnitude alone.

What χ^2 can reach here. χ^2/dof settles near 3,600, not 1, because the data contain subhalo substructure no smooth model describes: the *true* source through the *true* lens gives a median of 6,492 on 500 images, and a fitted smooth source can score below even that. The residual grows with brightness, 2,312, 5,514 and 40,025 across three signal-to-noise bins, which marks it as a systematic.

Which source representation the data support. The table is monotone: across nearly three orders of magnitude in the number of free source values, source error falls as that number falls across rows 2–5, which differ in nothing else. Row 1 sits outside that set: its 254^2 field is underdetermined seventeenfold and inflates the source tenfold at every prior strength across a hundredfold sweep, but it also used a fixed circular lens, worth 0.799 against 0.996 in arc-weighted correlation at fixed true source, so part of the inflation is the lens and not the free values. The two overdetermined hybrids do not escape it either, at nmse 0.0297 and 0.0336, on different analytic bases, only one improving on its own (Appendix F). The last row is a ceiling only because these sources are Sérsic profiles by construction, so the analytic prior is exactly correct; the table constrains this dataset, not learned source models in general. On real galaxies, where no analytic family is correct, a learned source is the only route left, and how many values it may carry is set by the roughly 3,878 informative pixels an image supplies.

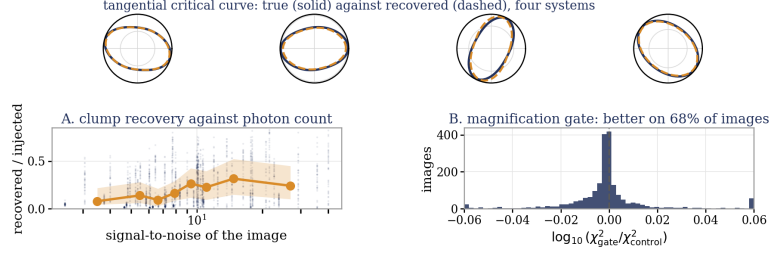


Figure 2: Top, the tangential critical curve of the recovered lens against the truth for four systems, none of which entered the fit. A, recovered over injected clump contrast against photon count. B, per-image χ^2 of the gated model relative to its control.

Is the recovered sub-pixel detail real? Reconstructing on a grid finer than the detector is meaningful only if the lens supplies the extra sampling: median tangential stretch is 2.86 against a radial 0.98, and the networks request 2.0 times. Warrant is not evidence, so we plant a Gaussian clump of 0.08 arcsec full width at half maximum, 0.76 detector pixels, at 15 percent of the source peak, and reconstruct with and without it under one noise realization, so noise and smooth source cancel in the difference. Over 2,400 injections across 60 images, none seen during training, a median 18 percent of the injected contrast returns, rising with photon count at partial rank correlation +0.33 (Figure 2A). It does not rise with local magnification, at -0.21 , but that comparison is weak by construction: the clump amplitude is fixed against the *global* source peak, so its local contrast grows with source-plane radius (Appendix F).

Building magnification into the prior does pay. A structural gate applies a sigmoid in $\log \mu$ to the network output before the forward model, permitting sub-pixel structure only where the lens delivered the resolution to constrain it. Against a control identical in every other respect it improves every source-plane metric on both seeds, taking nmse from 0.0336 and 0.0331 to 0.0321 and 0.0321, two to three times the seed-to-seed spread within the control (Appendix E). Data fidelity is essentially unchanged, the median χ^2/dof moving from 4,005 to 4,011, 0.15 percent, while per image the gated model has the lower χ^2 on 68 percent of the 2,000 images (Wilcoxon $p = 4 \times 10^{-30}$, Figure 2B). Weighting the correction penalty by magnification instead of gating moves nmse from 0.0302 to 0.0297, the same way.

Amortized inference as an initializer. A single network forward pass costs milliseconds against 0.94 s for a cold per-image fit and matches it on Einstein radius (+0.974) and half-light radius (+0.978), but fails on ellipticity, at +0.569 against +0.776: a conditional-mean shrinkage of a spin-2 quantity rather than a misorientation, with median phase error 9 degrees while the modulus is shrunk to 0.63 of the truth, flat across phase-error bins (Appendix F). We therefore initialize the fit with the network rather than replace it: with identical bounds and tolerances the warm fit reaches χ^2/dof 3,571 in a median 136 model evaluations and 0.41 s in one stage, against 3,649 in 475 and 0.94 s across four stages from cold, improving rank correlation on all seven parameters.

4 Limitations and outlook

Transfer outside the fitted lens family is untested. We report point estimates and no posterior, the largest gap and the one our results point at most directly, since the modulus shrinkage above is a bias a density head would report as a width. Magnification is derived from the recovered lens rather than measured independently, and all data are simulated from one instrument, so nothing here has met real galaxy morphology or a real point spread function.

Three directions follow. A density head over the fourteen parameters would report that shrinkage as a width. Real galaxy morphology, from a catalog or from a learned generative prior, would lift the ceiling of the last table row. Diffusion models have been applied here both as a conditional low- to high-resolution map [19] and as a source prior inside a forward model [16, 17, 18]; only the second needs no pairs, and our gate result suggests modulating its strength by local magnification. Finally, the lens is achromatic and both bands are present, so a two-band fit would constrain one lens with twice the data.

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A Further reconstructions

The forward chain drawn in Figure 1 and Figure 3 is the project’s own renderer run under NumPy from the frozen lens fit and the stored source map; it reproduces the PyTorch predictions used in every metric to 0.4 percent rms of the image peak.

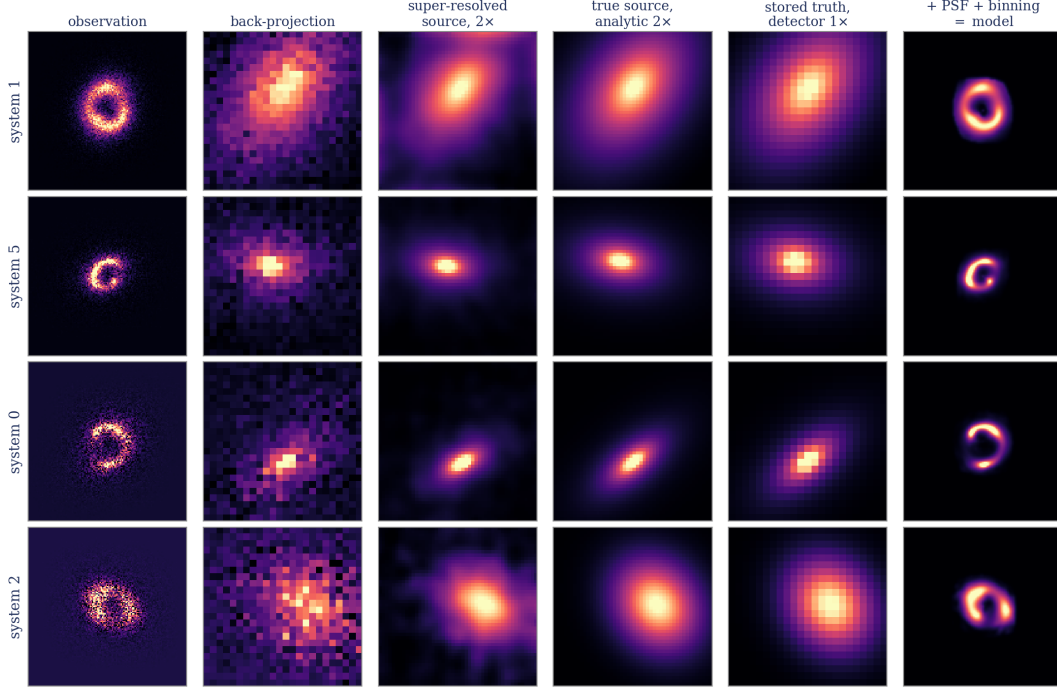


Figure 3: Four further validation systems through the pipeline of Figure 1, spanning the signal-to-noise range of the validation set. Panels and scaling follow that figure.

B Full parameter recovery

Per-image bounded nonlinear least squares (trust-region reflective rather than plain Levenberg-Marquardt, since several parameters carry hard bounds), initialized from the amortized network, $n = 2,000$ across the three dark matter classes.

parameter	fit median	true median	median abs. err	p90 abs. err	Spearman
θ_E (arcsec)	1.300	1.310	0.047	0.102	+0.973
β (arcsec)	0.287	0.306	0.030	0.089	+0.925
R_{serisic} (arcsec)	0.514	0.489	0.039	0.180	+0.947
n_{serisic}	1.183	1.003	0.174	0.901	+0.956
slope γ	2.015	2.050	0.068	0.233	+0.590
lens $ e $	0.203	0.219	0.031	0.107	+0.813
shear $ g $	0.048	0.033	0.014	0.052	+0.645

Source plane, scored against the stored source with the model convolved first: correlation 0.9918, size ratio 1.061, centroid error 0.353 pixels, peak ratio 0.863, nmse 0.0166. Median χ^2/dof is 3,571, with the plain background σ over the 6,361 pixel fit disk divided by $6,361 - 14$ for the fourteen free parameters. The radial slope is the least constrained parameter and sits close to isothermal.

C Instrument response

The stored unlensed array is the analytic source already convolved with the true kernel, so in Fourier space the kernel is the ratio of the two transforms. Naive division amplifies noise wherever the source transform is small, so we use the Wiener form, replacing $1/\hat{s}$ by $\hat{s}/(|\hat{s}|^2 + \lambda)$, and stack over 40 sources. This assumes nothing about the kernel's shape, which is the point: a parametric sweep over Gaussian widths bottoms out at a residual nmse of a few times 10^{-3} rather than at zero, so the true kernel is not Gaussian, and on arcs reaching peak-to-background ratios of 10^3 to 10^4 a one percent shape error is a 50σ per-pixel residual. The recovered kernel carries 22 percent of its flux beyond 0.25 arcsec against 13 percent for the Gaussian that best fits the same kernel in least squares.

D Resolution is not a rendering choice

The parametric source is continuous and can be sampled on any grid. Scored against the stored source rendered on the same grid, $n = 2,000$:

factor	scale (arcsec/px)	corr	nmse	size ratio
1×	0.1059	0.9843	0.0309	0.950
2×	0.0530	0.9835	0.0326	0.928
4×	0.0265	0.9832	0.0335	0.912
8×	0.0132	0.9830	0.0336	0.906

Correlation moves by 0.0013 across an eightfold range, so the sampling of the render is not what limits the reconstruction.

E Magnification-adaptive regularization

Both variants make the source prior depend on the local magnification of the recovered lens. The weaker one weights the curvature penalty on the 32^2 correction by $\mu^{1/2}$; it concentrates the correction where the lens magnifies, at rank correlation +0.50 between correction amplitude and $\log \mu$ against +0.32 for a uniform control, and moves source nmse from 0.0302 to 0.0297. The stronger one gates the 48^2 representation, passing the network output through a sigmoid in $\log \mu$ before the forward model.

run	epochs	corr	size ratio	peak ratio	nmse
control, seed 0	100	0.9829	1.048	0.930	0.0336
control, seed 1	50	0.9833	1.039	0.946	0.0331
gate, seed 0	100	0.9838	1.033	0.952	0.0321
gate, seed 1	50	0.9837	1.041	0.940	0.0321

The two gated runs agree to four decimals in nmse and both sit below both controls, whose own seed spread is 0.0005. The gate sits before the likelihood, so in principle the network can learn its inverse and undo it; the measured outcome is that it does not, and the constraint survives. The gate does not lower the median χ^2/dof , which is 4,011 against the control's 4,005, while the paired per-image comparison favors it on 68 percent of the same images. The two statistics disagree because they measure different things: the medians come from two separate distributions, whereas the paired test sees a skewed difference, a small gain on most images against a larger loss on a minority.

F Additional detail

The two hybrid rows. They are not added to the same base. The 48^2 residual sits on the per-image Sérsic and takes source nmse from 0.0166 to 0.0336. The 32^2 correction sits on a Sérsic the network predicts itself and improves that base from 0.0311 to 0.0297; the base is already worse than the fitted one, which is why the corrected row still lands above the analytic row of Table 1.

Informative pixels. Across 300 validation images in the three classes, the fit disk of 6,361 pixels contains a mean of 3,878 pixels above 3σ (p10 3,196, p90 4,545). This is the denominator of the “data per value” column in Table 1.

Spin-2 shrinkage. Modulus ratio by phase-error bin for the amortized network: 0.63, 0.64, 0.61 and 0.57 across 0 to 5, 5 to 15, 15 to 30 and 30 to 90 degrees, with counts 611, 726, 387 and 276. The per-image fit shows a median phase error of 3.3 degrees and a modulus ratio of 0.90 on the same images.

Injection detail. Recovered clump contrast correlates at +0.52 with the clump's source-plane radius, the strongest single predictor in the set, because the amplitude is fixed against the global source peak and not the local one. Whole-source reconstruction error falls from 0.34 in the $\mu = 1$

249 to 2 bin to 0.018 at $\mu = 8$ to 16, on per-bin image counts of 51, 60, 60, 58 and 24; that trend is
250 confounded with brightness, which is what the paired injection controls for. The clump is fixed at 15
251 percent of the global source peak wherever it lands, so the absolute contrast scale is arbitrary and
252 only relative trends are interpretable.